CLAIMS

What is claimed is:

1. A method for communicating data over a network, comprising:

- establishing a high-radix communication session between a first network device and a second network device over a communication network;
- dynamically defining a high-radix visual lexicon mapping communicative states to high-radix multi-spectral visual representations in accordance with a forward-secret parametric protocol;
- generating a natively optical data frame comprising high-radix visual elements structured according to said high-radix visual lexicon;
- transmitting said natively optical data frame across said communication network to the second network device; and
- systematically rotating said high-radix visual lexicon across subsequent transmission intervals according to said forward-secret parametric protocol, such that identical communicative states are transmitted as deterministically different visual representations, thereby destroying statistical language footprints across the communication network.

2. The method of claim 1, wherein the first network device comprises an optically-native central processing unit generating wide data words directly in native multi-spectral optical states, and wherein the natively optical data frame is routed across the communication network by a pixel-based optical network router without serial-binary deserialization.

3. The method of claim 1, wherein the first network device comprises a legacy serial-binary computing device, and wherein said generating the natively optical data frame comprises executing a software or hardware retrofit codec translating an incoming serialized binary byte stream into said high-radix visual elements.

4. The method of claim 1, further comprising:

- receiving said natively optical data frame at a secure operations facility; and
- natively recording said natively optical data frame directly onto a high-density immutable physical optical media substrate in real-time, bypassing serial-binary down-conversion.

5. The method of claim 1, wherein the high-radix visual elements are grouped into logical n-grams spanning a plurality of adjacent physical optical channels, such that the logical n-grams represent a combined state space equal to the product of states of the constituent physical optical channels.

6. An end-to-end high-radix network system, comprising:

- a transmitting terminal comprising an optically-native central processing unit and a parametric lexicon engine;
- an optical network comprising at least one pixel-based optical router configured to switch high-radix visual frames without serialization;
- a destination terminal comprising a native optical receiver and a synchronized parametric lexicon engine; and
- an immutable optical archival engine configured to execute a real-time native optical dump of network traffic directly onto high-density physical optical media.